BINDING OF A THYROXINE MIMETIC TO TRANSTHYRETIN

M. HASHIM AKBAR (2307943)

SUPERVISOR: STEVE WOODS

BIOTECHNOLODGY RESEARCH SKILL PROJECT

ABSTRACT:

Folding errors and deposits of the TTR protein are hallmarks of the debilitating disease transthyretin amyloidosis (TTR). Preserving the native TTR tetramer to prevent its dissociation into amyloidogenic monomers is a crucial therapeutic approach. The aim of this work was to use X-ray crystallography to elucidate the molecular basis of TTR stabilization by a novel chlorinated triazole ligand. Through iterative modeling (Coot), refinement (REFMAC5), and molecular replacement (Phaser), the excellent quality structure of TTR in combination with the ligand was successfully identified. The results indicate that a divalent chlorinated triazole ligand possesses pseudo-irreversible binding affinity and forms a strong bond at the dimer-dimer interface at the center of the TTR tetramer. In contrast to monovalent thyroxine mimetics, this specific binding mode effectively locks the tetramer in a stable conformation, thus limiting its dissociation and subsequent amyloid fibril formation. This structural knowledge enables the rational development of future TTR stabilizers.

INTRODUCTION:

The 55 kDa tetrameric transport protein known as transthyretin (TTR) is made up of four identical subunits, each containing 127 amino acids. which is produced in the retinal pigment epithelium, liver, and choroid plexus before being released into the bloodstream, brain fluid, and eye, respectively (Li, 2011).

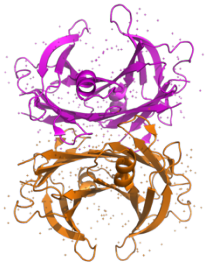

Figure 1: The transthyretin (TTR) homo-tetramer's general structure, showing its four identical subunits (rendered with PyMOL)

TTR could split into full-length and fragmented monomers in vivo, which then combine to form amyloid fibrils. These fibrils build up extracellularly in organs and tissues, primarily the heart and peripheral nerves. This leads to transthyretin (ATTR) amyloidosis, a dangerous, progressive condition that exhibits significant heterogeneity, with individual variances in symptom presentation, clinical manifestation, and disease risk (Rapezzi, 2010). ATTR amyloidosis comes in two forms: wild-type (ATTR-wt) and inherited (ATTR-v; v for variant). These seem to have significant physio-pathological pathways in common (Marcoux, 2015). Adult ATTR amyloidosis that is dominantly inherited can result from polymorphisms in the TTR gene. There are clinical and regional distinctions among early-

onset and late-onset types of the disease, and this may happen starting around age 30 but is more prevalent beyond age 50 (Koike, 2002). Within ATTR-wt, the typical protein usually collects in the heart, leading to an aging-related restrictive, pseudo-hypertrophic cardiomyopathy, males are more prone to ATTR-wt (Rapezzi, 2010).

There are now two primary treatment approaches that work well for ATTR stabilization. A nonsteroidal anti-inflammatory drug with TTR stabilizing qualities is diflunisal. To stop the rate-limiting detachment of TTR and its resulting fold errors form amyloidogenic monomers and oligomers, it binds to the T4-binding regions at the dimer-dimer junction on the transthyretin tetramer. Hypertension, fluid retention, renal failure, and gastrointestinal bleeding are among the side effects linked to diflunisal (Cheng, 2021). Like diflunisal, Tafamidis (Vyndamax/Vyndaqel, Pfizer) is an oral bioavailable benzoxazole derivative that binds to the T4-binding sites to stabilize the TTR tetramer but does not have nonsteroidal anti-inflammatory properties (Cheng, 2021).

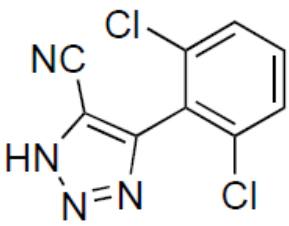

Figure 2 : Two-dimensional chemical structure of the chlorinated triazole ligand ($C_8H_3Cl_2N_5$)..

In this study, a novel chlorinated triazole candidate—a bivalent ligand intended to provide a clear benefit—is investigated. This chemical is thought to achieve a much tighter, pseudo-irreversible binding affinity by attaching a chlorinated triazole molecule to an MDS84 unit and a tafamidis derivative. In contrast to traditional monovalent inhibitors, this novel bivalent design is suggested to interact with the TTR tetramer across a wider network of contacts, offering a more reliable method of preventing TTR fragmentation into amyloidogenic fibrils.

METHODOLOGY:

To do macromolecular crystallographic analysis, the CCP4 suite was installed. The ligand structure was sketched and exported using Biovia Draw. To begin creating the model, a transthyretin (TTR) search model (PDB ID: 2G5U) was taken from the Protein Data Bank.

Utilizing the known TTR structure, Phaser was used to carry out molecular replacement and produce preliminary phase estimates. Coot was used to view the final structure and electron density maps (2Fo–Fc and Fo–Fc). Where the solvent presence was supported by electron density, water molecules were introduced.

To preserve stereochemical precision, the first refinement was carried out using Refmac5 with limited parameters (such as restrictions on bond lengths, angles, planar groups, and thermal factors). Weakly dense or geometrically inconsistent regions were manually refined in real space in Coot. To enhance overall model quality, iterative rounds of Refmac5 refining and Coot model rebuilding were carried out.

Using Biovia Draw, a chlorinated triazole ligand ($C_8H_3Cl_2N_5$) was created and saved as a.mol file. AceDRG was used to convert the structure into.pdb and.cif formats and produced the restraint dictionaries needed for refining. Based on the Fo–Fc positive density, the ligand was physically docked into the site of action and integrated into the Coot map. To get the best fit, occupancy and B-factors were manually changed.

Refmac5 was used to refine the entire protein–ligand complex one last time. Geometry checks, visual map agreement, and the lack of steric conflicts were the main focuses of structural validation. PyMOL was used to view the finished model, and graphics were produced to emphasize active site interactions and ligand binding.

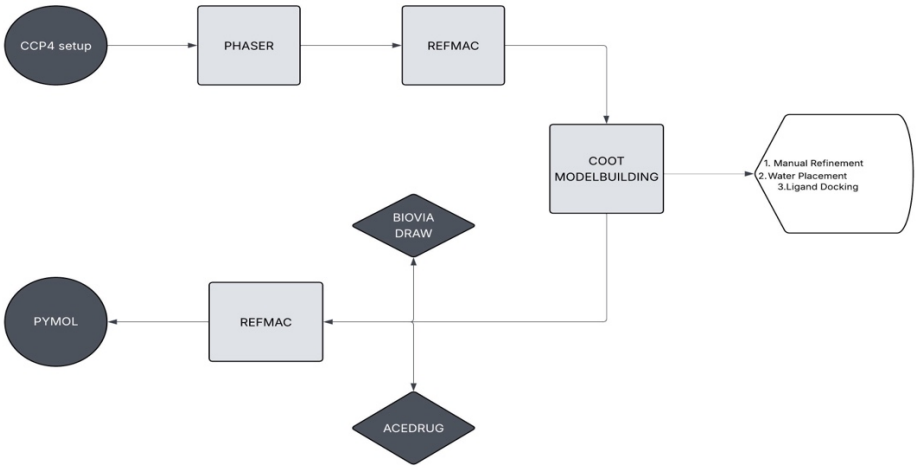

Figure 3: Workflow chart diagram showing the TTR-chlorinated triazole complex's crystallographic analysis approach (created using LucidChart)

RESULTS:

For molecular replacement, Phaser was used (A. J. McCoy, 2007). A clear solution with a Rotate Function LLG of 35.29 (RFZ = 7.57).

```
   Rotation Function Table: ensemble1
   ----------------------------------
   (Z-scores from Fast Rotation Function)
   #SET        Top    (Z)      Second    (Z)       Third    (Z)
   1         35.29   7.57       27.03   7.04        ---    ---


---------------
FINAL SELECTION
---------------


   Select by Percentage of Top value: 75%
   Top RF = 35.294
   Purge RF mean = -83.535
   Number of sets stored before final selection = 1
   Number of solutions stored before final selection = 2
   Number of sets stored (deleted) after final selection = 1 (0)
   Number of solutions stored (deleted) after final selection = 2 (0)
   Percent used for purge = 60.000%
      Includes deep search down percent = 15%
      Number of solutions stored above (below) deep threshold = 2 (0)

   Rotation Function Final Selection Table
   ---------------------------------------
   Rotation list length by SET
   SET#  Start Final Deleted Set (*)   Deep: Start Final Deleted Set (*)
      1  2     2           -                 2     2           -
   ALL   2     2                             2     2
```

Figure 4: Rotation function results from Phaser, displaying the top solutions and their Z-scores. (Source: Phaser log file)

Another Translation Function LLG of 123.32 (TFZ = 11.59) was found using the 2pab.pdb model (ensemble1) as a search model.

```
   Translation Function Table
   --------------------------
   SET ROT*deep   Top   (Z)      Second   (Z)      Third   (Z) SpaceGroup  Ensemble
     1   1      123.3 11.59          -     -          -     - P 21 21 2 ensemble
     1   2       38.1  9.36          -     -          -     - P 21 21 2 ensemble
   --- ---

---------------
FINAL SELECTION
---------------

   LLG will be used for purge, not FSS
   Top TF  = 123.32
   Top TFZ = 11.59
   Mean TF = -346.12
   Percent used for purge = 75%
   Cutoff for acceptance = 6.0
      TFZ used for final selection = 8
         Number of solutions over TF final cutoff  = 2
         Number of solutions over TFZ final cutoff = 2
         Number of solutions over TF & TFZ cutoff  = 2
   Number of solutions stored before final selection = 2
   Number of solutions stored (deleted) after final selection = 2 (0)


$TABLE : Translation Function Component #1 (ensemble1):
$GRAPHS
:TF Number vs LL-gain:AUTO:1,2: :TF Number vs Z-Score:AUTO:1,3:
$$
Number LLG Z-Score
$$ loggraph $$
1     123.32 11.59
2      38.14  9.36
$$
```

Figure 5: Phaser's translation function results, displaying the best answers together with their Z scores. (Source : Phaser log file)

```
-------
RESULTS
-------

   Refinement Table (Sorted)
   -------------------------
   Refinement to full resolution
   #out =#out #in =T (Start LLG Rval TFZ) (Refined LLG Rval TFZ==) SpaceGroup  Cntrst
   Top1 ---   1          2105.6 45.8 13.5       2616.0 45.1 42.1   P 21 21 2     n/a

   Refinement Table (Variance Ranges)
   ----------------------------------
   Range of delta-VRMS and VRMS given over current solution list (1 solution(s))
   Ensemble        Model#   RMS   Delta-VRMS min/max  (VRMS min/max)
   ensemble1            1   0.419   -0.075/-0.075    ( 0.317/ 0.317 )
```

Figure 6: Refinement results from Phaser, detailing the improvement in LLG, R-value, and TFZ after refinement of the molecular replacement solution to full resolution. (Source: Phaser log file)

The solution was refined further to the full resolution of 1.49 Å after initial rigid-body refinement. This resulted in a final LLG of 2616.0 and an extremely notable TFZ of 42.1, with the R-factor stabilizing at 45.1%.

Using REFMAC5, the first round of constrained refinement was performed on the original molecular replacement solution(Vagin, 2004). The initial R-factor and R-free for this 10-cycle preliminary refinement were 0.4317 and 0.4293, respectively. By the end of this first stage, the R-free had reached 0.3201 and the R-factor was converging to 0.2901.

```
$TABLE: Rfactor analysis, stats vs cycle  :
$GRAPHS:<Rfactor> vs cycle :N:1,2,3:
:FOM vs cycle :N:1,4:
:-LL vs cycle :N:1,5:
:-LLfree vs cycle :N:1,6:
:Geometry vs cycle:N:1,7,8,9,10,11:
$$
    Ncyc    Rfact    Rfree     FOM      -LL     -LLfree  rmsBOND  zBOND rmsANGL  zANGL rmsCHIRAL $$
$$
       0   0.4178   0.4159   0.536      192760.   10206.0   0.0172  1.372   3.982  2.340   0.398
       1   0.3773   0.3925   0.579      189588.   10096.1   0.0080  0.662   2.449  1.365   0.177
       2   0.3503   0.3698   0.625      186451.    9957.6   0.0080  0.630   2.156  1.165   0.127
       3   0.3292   0.3532   0.661      183788.    9838.8   0.0083  0.641   2.072  1.090   0.116
       4   0.3125   0.3403   0.687      181638.    9744.6   0.0085  0.658   2.057  1.061   0.113
       5   0.2991   0.3301   0.706      179938.    9670.8   0.0086  0.667   2.048  1.044   0.112
       6   0.2912   0.3239   0.718      178902.    9624.3   0.0088  0.683   2.054  1.045   0.113
       7   0.2847   0.3183   0.727      178111.    9589.5   0.0088  0.688   2.051  1.041   0.113
       8   0.2793   0.3133   0.736      177397.    9556.7   0.0088  0.691   2.046  1.035   0.114
       9   0.2746   0.3084   0.743      176730.    9525.2   0.0088  0.694   2.040  1.030   0.114
      10   0.2709   0.3041   0.748      176271.    9503.5   0.0088  0.688   2.036  1.027   0.114
 $$
 $TEXT:Result: $$ Final results $$
                      Initial    Final
           R factor    0.4178   0.2709
             R free    0.4159   0.3041
     Rms BondLength    0.0172   0.0088
      Rms BondAngle    3.9825   2.0364
     Rms ChirVolume    0.3985   0.1144
 $$
<B><FONT COLOR="#FF0000">
 Refmac:  End of Refmac_5.8.0430
Times: User:      14.0s System:    0.1s Elapsed:     0:14
```

Figure 7: REFMAC5 refinement statistics that display geometric parameters, R-factor, and R-free values as a function of refinement cycle. (Source: REFMAC5 log file)

In Coot, extensive manual model construction and mistake correction were carried out after this initial improvement (Cowtan, 2010). making use of electron density maps for 2Fo-Fc and Fo-Fc. In this repeated approach, the chlorinated triazole ligand was precisely fitted, side-chain conformations and backbone were adjusted, and misunderstandings were fixed.

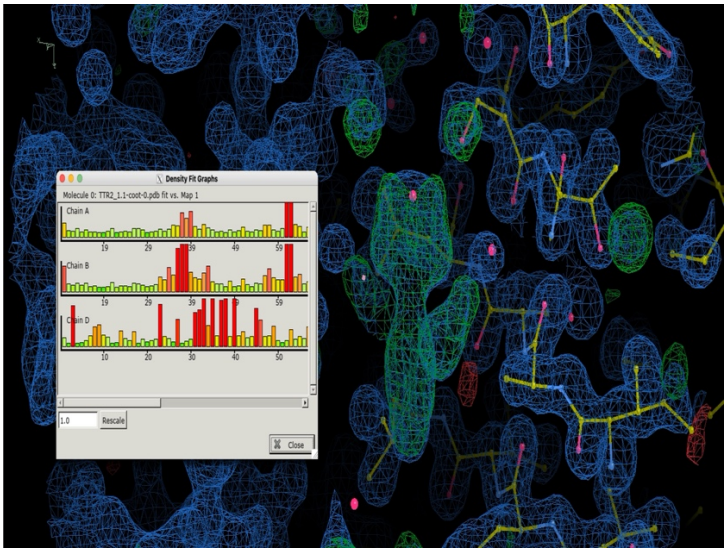

Figure 8: Initial density fit analysis and electron density maps in Coot show that the model must be rebuilt. (Source: Coot visualization)

For instance, Figure 8 depicts an area where manual modifications have not yet been completed, and the density fit graph indicates the amount of construction that must be performed within the 2Fo-Fc electron density at (contoured at 1.49 σ).

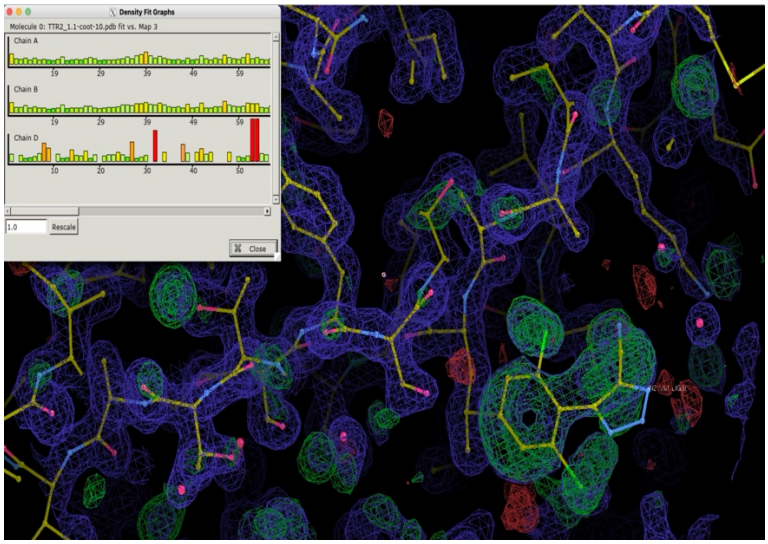

Figure 9: The ligand and the completed model are both easily discernible within the positive electron density. Comparing "The density fit graphs" inset to the original model, the areas of poor fit (red bars) have significantly decreased. (Source: Coot visualization)

Additionally, as shown in Figure 9, flexible loop regions—which were frequently difficult to accommodate in early models—were meticulously adjusted to match the measured electron density. Importantly, the Fo-Fc differential map (favourable density in green, unfavorable in red, both contoured at 1.49 σ) confirmed a good fit and negligible residual density, and the chlorinated triazole ligand was visibly positioned within an established 2Fo-Fc electron density (contoured at 1.49 σ).

```
$TABLE: Rfactor analysis, stats vs cycle  :
$GRAPHS:<Rfactor> vs cycle :N:1,2,3:
:FOM vs cycle :N:1,4:
:-LL vs cycle :N:1,5:
:-LLfree vs cycle :N:1,6:
:Geometry vs cycle:N:1,7,8,9,10,11:
$$
    Ncyc    Rfact    Rfree     FOM      -LL     -LLfree  rmsBOND  zBOND rmsANGL  zANGL rmsCHIRAL $$
$$
       0   0.2356   0.2702   0.796      172297.    9334.0   0.0105  0.844   1.997  1.036   0.104
       1   0.2341   0.2678   0.799      172024.    9312.0   0.0101  0.806   1.983  1.026   0.104
       2   0.2334   0.2676   0.800      171936.    9309.1   0.0101  0.799   1.982  1.025   0.104
       3   0.2330   0.2673   0.800      171878.    9305.7   0.0101  0.798   1.989  1.027   0.104
       4   0.2326   0.2672   0.801      171841.    9303.8   0.0102  0.800   1.998  1.031   0.105
       5   0.2324   0.2668   0.802      171805.    9301.3   0.0103  0.806   2.006  1.036   0.105
       6   0.2322   0.2665   0.802      171782.    9299.8   0.0104  0.812   2.014  1.040   0.105
       7   0.2321   0.2658   0.803      171763.    9298.3   0.0105  0.818   2.021  1.044   0.105
       8   0.2320   0.2661   0.803      171749.    9298.0   0.0106  0.824   2.026  1.048   0.106
       9   0.2317   0.2659   0.803      171733.    9297.9   0.0107  0.828   2.031  1.050   0.106
      10   0.2316   0.2658   0.803      171727.    9297.7   0.0107  0.827   2.026  1.050   0.106
 $$
 $TEXT:Result: $$ Final results $$
                      Initial    Final
           R factor    0.2356   0.2316
             R free    0.2702   0.2658
     Rms BondLength    0.0105   0.0107
      Rms BondAngle    1.9973   2.0265
     Rms ChirVolume    0.1039   0.1059
 $$
<B><FONT COLOR="#FF0000">
 Refmac:  End of Refmac_5.8.0430
Times: User:      13.2s System:    0.1s Elapsed:     0:14
```

Figure 10: A well-refined model is indicated by the final REFMAC5 refinement statistics, which display the conversion of R-Factor and R free values over 10 cycles as well as the RMSD values for bond lengths, bond angles, and chiral volumes. (Source: REFMAC5 log file)

The model underwent a final round of constrained refining using REFMAC5 after undergoing a thorough manual rebuild in Coot. The R-factor and R-free values for this post-model construction refinement, which also lasted 10 cycles, were noticeably higher at 0.2356 and 0.2702, respectively. The refinement process resolved to a concluding R-factor of 0.2316 and an R-free of 0.2658, indicating that the combination of manual and computational refinement efforts significantly improved the quality of the model.

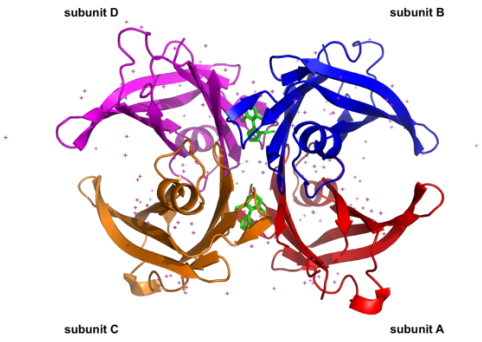

Figure 11: Whole structure of the transthyretin (TTR) homo tetramer in association with the bound ligand (green sticks) (ribbon representation, with distinct colors for each monomer). The ligand is located inside the central binding channel, while the pink spheres represent the surrounding water molecules. (Rendered Using PyMOL)

The study of the improved TTR-ligand complex, which builds on the structural discoveries, provides a clear illustration of the molecular foundation for stabilization. As illustrated in Figure 11 the chlorinated triazole ligand binds precisely at the dimer-dimer interface, in the central channel of the TTR Hom tetramer. The tetrameric structure is essentially locked into a stable shape by this crucial interaction, which strengthens the non-covalent bonds between the four monomers. The ligand provides a therapeutic method against transthyretin amyloidosis by directly inhibiting the resulting misfolding and aggregation events that result in the development of amyloid fibrils by stopping the restricting fragmentation of the TTR tetramer into unstable monomers.

DISCUSSION:

The ligand binds to the homo-tetrameric protein asymmetrically, consistent with crystallographic data, with two different orientations observed at the otherwise identical binding sites. Despite being physically identical in the apo state, this asymmetry challenges the widely accepted concept of two-fold symmetry in such complexes and suggests that the binding sites become functionally non-equivalent upon ligand engagement.

The observed divergence between the binding modes suggests negative cooperativity. If the first ligand binds with high affinity to a single binding site, this can induce a conformational shift that alters the structural composition of the second binding site and reduces the binding capacity for the ligand. Despite previous proposals, the theory of negative cooperativity often lacked direct structural support. Because the two ligand binding processes are not identical and the conformational state of the protein varies

between the two binding events, our results provide strong structural evidence for this process.

Furthermore, the stacking of crystals is significantly influenced by this asymmetry. As with the stacking of identical building blocks, the formation of a crystal lattice in traditional crystallography is usually based on repeating elements of uniform shape. However, a theoretical question arises when the ligand within a single tetramer exists in two different orientations: How does the crystal lattice resolve this heterogeneity? If the tetramers were truly symmetric with randomly oriented ligands, any clear ligand orientation would be obscured by the average electron density of the resulting crystal. Our structure, which shows a consistent ligand arrangement in each tetramer of the crystal, instead suggests an unpredictable, directional arrangement.

This finding suggests that the tetramer exhibits an intrinsic asymmetry sufficient to distinguish one end of the group from the other. This asymmetry is likely caused by the initial high-affinity ligand binding. During crystal formation, this structural difference allows the lattice to recognize and uniformly position each tetramer. Our results thus not only confirm that the binding sites are not equivalent but also suggest a process by which this asymmetry governs ordered crystal formation and binding cooperativity.

Bibliography:

A. J. McCoy, L. C. (2007). *Phaser crystallographic software*. Retrieved from Phaser.

Cheng, J. M. (2021). ATTR Amyloidosis: Current and Emerging Management Strategies. *JACC: CardioOncology*, 488-505.

Cowtan, P. E. (2010). Features and Development of Coot. *Acta Crystallographica Section D - Biological Crystallography*, 486-501.

Koike, H. a.-i.-i. (2002). Type I (transthyretin Met30) familial amyloid polyneuropathy in Japan: early-vs late-onset form. *Archives of neurology, 59*(11), 1771-1776.

Li, X. a. (2011). Transthyretin and the brain re-visited: is neuronal synthesis of transthyretin protective in Alzheimer's disease? *molecular neurodegeneration*, 1-17.

Marcoux, J. a.-C. (2015). A novel mechano-enzymatic cleavage mechanism underlies transthyretin amyloidogenesis. *EMBO molecular medicine*, 1337-1349.

Rapezzi, C. a. (2010). Transthyretin-related amyloidoses and the heart: a clinical overview. *Nature Reviews Cardiology, 7*(7), 398-408.

Vagin, A. S. (2004). organisation of prior chemical knowledge and guidelines for its use. *REFMAC5 dictionary*, 2284-2295.